

Influence Mechanism of Fluorine on the Preparation of Vaterite by Phosphogypsum Direct Mineralization: Insights from Experimental and Density Functional Theory Analyses

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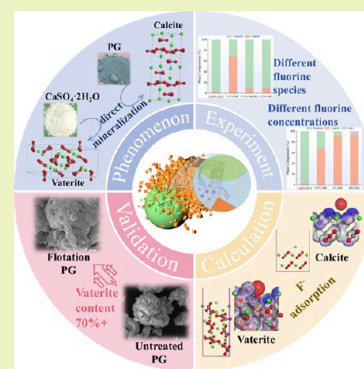
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ABSTRACT: Phosphogypsum (PG), a byproduct of the phosphate fertilizer industry, is a promising candidate for CO₂ mineralization due to its high calcium content. However, the direct mineralization of PG typically yields calcite rather than the metastable vaterite phase of CaCO₃, which holds superior industrial value. This study investigates the influence of impurities in PG on vaterite formation through experimental and density functional theory (DFT) approaches. Through the analysis of diverse impurity types in PG, fluorine was identified as the dominant element influencing vaterite formation. Mineral carbonation experiments were conducted using pure CaSO₄·2H₂O and PG with varying fluorine species (NaF, Na₂SiF₆, and CaF₂) and concentrations. It revealed that soluble fluorine significantly promotes calcite formation, even at a low concentration of 0.1 wt %, while sparingly soluble fluorides exhibit minimal effects. DFT calculations demonstrated that F[−] adsorption on vaterite (110) surfaces involves strong chemical interactions, destabilizing the structure and facilitating its transformation to calcite. In that case, a selective flotation process was used to remove soluble fluorine in PG, indicating that flotation pretreatment effectively reduced fluorine content in PG from 0.62 to 0.13%. Flotation PG enables the synthesis of high-purity vaterite with 70% content via direct mineralization. These findings elucidate the critical role of fluorine in governing CaCO₃ polymorphism and provide a practical pathway for enhancing the value of PG-derived products through impurity control.

KEYWORDS: phosphogypsum, direct mineralization, vaterite, fluorine, DFT



1. INTRODUCTION

Phosphogypsum (PG), a byproduct of the phosphate fertilizer industry, has accumulated globally in stockpiles exceeding 6 billion tons. This massive stockpiling occupies extensive land resources while posing severe safety hazards, particularly the environmental pollution impact of F and P.^{1–3} The main component of PG is CaSO₄·2H₂O, in which the content of CaO exceeds 30%. Due to its substantial content of active Ca, PG has enormous potential for CO₂ sequestration.^{4–6} The carbonation process utilizing PG presents a dual environmental advantage: it enables sustainable reuse of industrial waste while concurrently achieving a CO₂ emission reduction through mineralization. Furthermore, the generation of high-purity CaCO₃ products significantly enhances the valorization potential of PG, providing a strategic pathway for upgrading industrial by-products into a commercially viable material.^{7,8}

It is well-known that CaCO₃ has three anhydrous polymorphs with distinct physicochemical properties: calcite, aragonite, and vaterite.^{9,10} Extensive research has established that vaterite exhibits superior functional characteristics, including enhanced porosity,¹¹ increased specific surface area,¹² improved biodegradability,¹³ heightened hydrophilicity,¹⁴ and relatively low density.¹⁵ These exceptional properties render vaterite partic-

ularly advantageous for advanced material applications.^{16,17} Currently, high-value-added CaCO₃ with the vaterite phase obtained from PG was mainly prepared by the indirect mineral carbonation method.^{18,19} Experimental evidence indicates that controlled reactions between PG, NH₄OH, and NaCl can generate mixed-phase CaCO₃ products containing both calcite and vaterite components. Notably, systematic modulation of NaCl cycling parameters enables precise regulation of crystalline phase composition.²⁰

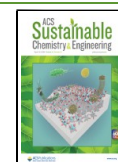
Compared to indirect mineralization, direct carbonation methods simplify the procedure by eliminating intermediate calcium extraction steps, thereby circumventing challenges associated with extractant recovery and recycling.²¹ However, due to the complex composition of impurities in PG, the quality of calcium carbonate obtained by the one-step method is inferior

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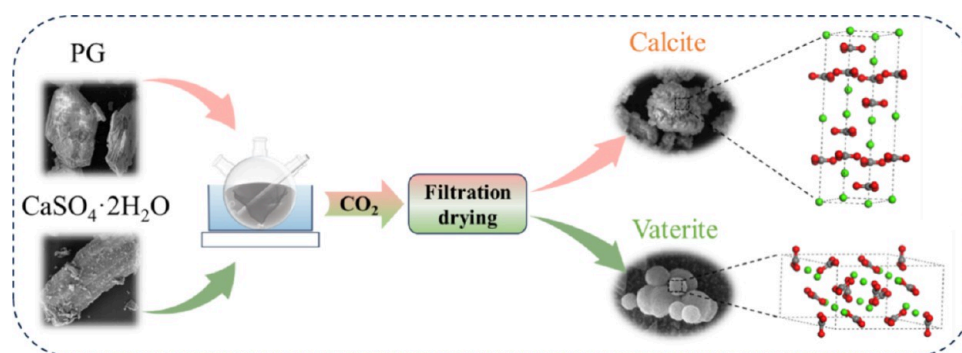


Figure 1. Different crystal forms of CaCO_3 obtained from the mineralization of different raw materials.

to that by indirect mineralization. Current research on direct mineralization of PG has predominantly focused on enhancing reaction efficiency,^{22,23} while investigation into the influence of intrinsic PG impurities on the crystal phases of mineralization products remains notably absent. As shown in Figure 1, in our previous study, CaCO_3 prepared by PG mineral carbonation was all calcite.^{24,25} At the same time, the carbonated products obtained by the $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ reagent under the same experimental conditions were all vaterite. Therefore, it is crucially important to understand the influence mechanism of impurities in PG on the crystalline structure of CaCO_3 to improve the quality of the PG direct mineralization products. While density functional theory (DFT) is a commonly used method to investigate additive adsorption on vaterite and calcite surfaces,^{26–28} no detailed studies have described how impurities in PG influence the polymorphic formation of CaCO_3 .

This study systematically investigates the impacts of various impurity components on the crystalline architecture of CaCO_3 products by PG direct mineralization. Complementary DFT simulations elucidate the underlying mechanisms of impurity influence on the metastable vaterite polymorph formation. This mechanism was further validated through direct mineralization employing flotation-pretreated PG with diminished fluorine content. The integrated experimental–calculation–validation approach establishes a theoretical framework for enhancing PG valorization while optimizing an energy-efficient vaterite synthesis approach.

2. MATERIALS AND METHODS

2.1. Materials

Untreated and flotation PG used in this study were obtained from Yichang, China, and were dried at 85 °C for 24 h to remove the adsorbed water for further experiments. $\text{NH}_3 \cdot \text{H}_2\text{O}$ (25–28%) was bought from Sinopharm Chemical Reagent Co., Ltd. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, SiO_2 , NaF, apatite, CaF_2 , and Na_2SiF_6 were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. and were of analytical reagent grade. The purity of the CO_2 gas used during the PG mineral carbonation is 99.9%. All solutions used in this work were prepared with deionized water.

2.2. Experimental Methods

First, 20 g of the solid sample (pure $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, untreated PG, or flotation PG), 200 mL of the mixed solution of ammonia (20 mL), and deionized water were placed in a three-port flask. When exploring the impact of impurity components, a specific quantity of impurity reagent should be added simultaneously. Then, the three-neck flask was placed in a constant temperature water bath to ensure that the reaction temperature remained at 25 °C. Meanwhile, the suspension was stirred at 200 rpm to make sure it was mixed evenly. After 10 min, the CO_2 gas was pumped into the continuously stirred suspension to start the

carbonation reaction. After 45 min of reaction, the final carbonated products were obtained through washing, filtering, and drying at 85 °C.

2.3. Analysis Methods

The chemical composition of untreated and flotation PG was determined by X-ray fluorescence (XRF; ZSXPrimus IV, Rigaku, JPN). The crystalline phase of carbonated products under various component conditions was analyzed using X-ray diffraction (XRD; SmartLab SE, Rigaku, JPN). Field-emission scanning electron microscopy (SEM-EDS; Apreo S HiVac, ThermoFisher, USA) was used to characterize the micromorphology and elemental composition of the different PG and carbonated products. The surface elements of untreated PG were analyzed by X-ray photoelectron spectroscopy (XPS; AXIS SUPRA+, Shimadzu, and JPN).

2.4. Measurement Methods of the Relative Content of Vaterite and Calcite

The relative content of vaterite and calcite can be calculated using the data from the XRD results through eqs 1 and 2:

$$X_V = \frac{7.691I_V^{110}}{I_C^{104} + 7.691I_V^{110}} \quad (1)$$

$$X_C = 1 - X_V \quad (2)$$

where X_V and X_C are the relative contents of vaterite and calcite in carbonated products, respectively. The spectral line integral intensities of vaterite(110) and calcite(104) in the XRD pattern of the carbonated products were denoted as I_V and I_C .²⁹

2.5. DFT Calculation Methods

All calculations in this investigation were conducted by employing the CASTEP module within Materials Studio 2020.³⁰ DFT calculations focused on F^- adsorption behaviors across calcite(104) and vaterite(110) surface configurations.^{31,32} Systematic convergence testing of plane-wave cutoff energies and k -point grids across different surface systems enabled optimization of calculation parameters: cutoff energies of 570 and 340 eV were implemented alongside k -point meshes of $2 \times 3 \times 1$ and $2 \times 2 \times 1$, respectively. This step balances numerical accuracy with computational efficiency.^{33,34} Consistent with previous studies, while the adsorbent layer quantity may influence adsorption energy quantification, it exhibits negligible effects on adsorbent geometrical configuration or adsorption site conformations.^{35,36} To establish an optimal balance between numerical precision and computational tractability for F^- adsorption modeling, a bilayer periodic slab model was adopted for all surfaces. Adjacent slabs were separated by a 10 Å vacuum layer to eliminate interslab interactions. During geometry optimization procedures, atomic positions in the bottom layers remained constrained, whereas the upper-layer atoms underwent full relaxation.

The surface energy (γ) of different surface slabs is defined and calculated as follows:

$$\gamma = \frac{E_s - nE_b}{2A} \quad (3)$$

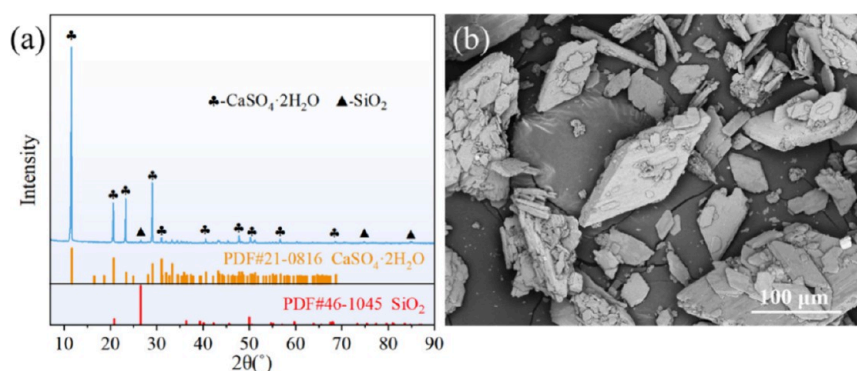


Figure 2. (a) XRD pattern of PG and (b) SEM image of PG.

Table 1. Chemical Composition of Untreated PG (wt %)

CaO	SO ₃	Al ₂ O ₃	SiO ₂	P ₂ O ₅	Fe ₂ O ₃	F	H ₂ O ^a	others
32.19	40.62	0.49	5.18	0.79	0.25	0.62	19.25	0.61

^aThe H₂O content specified here denotes the water of crystallization.

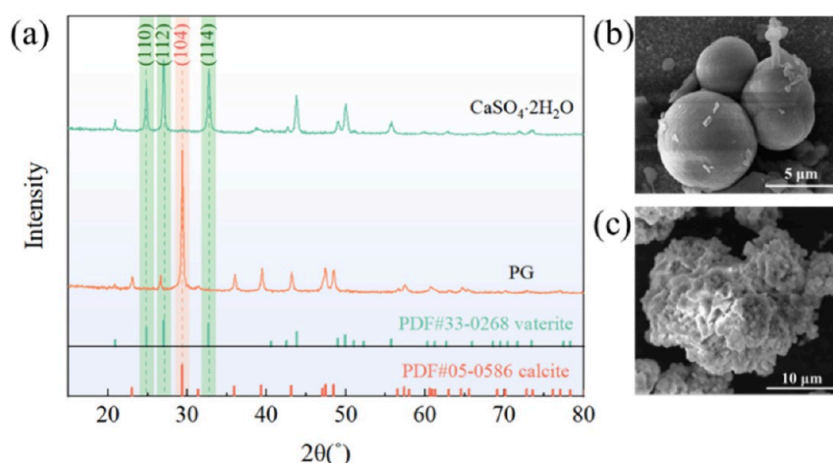


Figure 3. (a) XRD patterns of the carbonated products of untreated PG and pure CaSO₄·2H₂O reagents. SEM image of the carbonated products of (b) pure CaSO₄·2H₂O reagents and (c) untreated PG.

where E_s is the energy of a slab containing n formula units and E_b is the total energy per formula unit of CaCO₃. A is the area of the slab surface, and the factor of 2 reflects that there are two surfaces for each slab.³⁷

The adsorption energy (E_{ads}) is the variation in the total energy between species before and after adsorption. It can judge the possibility of adsorption and is calculated as follows:

$$E_{\text{ads}} = E_{(\text{A/surf})} - (E_{\text{A}} + E_{\text{surf}}) \quad (4)$$

where $E_{(\text{A/surf})}$ is the overall model energy of F[−] adsorption on different crystal forms of CaCO₃ surfaces in the equilibrium state. E_{A} and E_{surf} are the total energies of the free F[−] and original CaCO₃ surfaces, respectively.³⁸

3. RESULTS AND DISCUSSION

3.1. Impact of Impurity Types on the Preparation of Vaterite

Figure 2a shows the XRD pattern of untreated PG. The predominant crystalline phase is CaSO₄·2H₂O, along with a minor amount of SiO₂. From Figure 2b, it can be seen that the microstructure of PG is primarily composed of aggregated flaky parallelogram-shaped grains, accompanied by irregularly shaped fine particles.

As the chemical compositions of PG are shown in Table 1, the main compositions of PG were determined to be 32.19% (CaO)

and 40.62% (SO₃). It can be calculated that the component of CaSO₄·2H₂O in PG is about 92.06%.

Considering that the main component of PG is CaSO₄·2H₂O, mineral carbonation experiments were conducted using both pure reagents and untreated PG, respectively. The XRD patterns and SEM images of both carbonated products are shown in Figure 3. Under the same experimental conditions, CaCO₃ obtained from pure CaSO₄·2H₂O reagent exhibits a vaterite structure, while CaCO₃ synthesized from untreated PG has the structure of calcite. The vaterite-type CaCO₃ crystal displays a smooth, dispersed spherical morphology with a diameter of about 5 μm. The XRD pattern reveals that the diffraction peaks corresponding to the (110), (112), and (114) crystal planes exhibit higher intensities. Conversely, the calcite-type CaCO₃ obtained from untreated PG consists of small cubic particles with an average particle size of 65.82 μm, and its XRD pattern shows the highest peak intensity for the (104) plane.

The main difference of untreated PG and pure CaSO₄·2H₂O reagents is the presence of impurities in PG. To elucidate the formation mechanisms of the two distinct crystalline CaCO₃ structures, it is crucial to identify and characterize the forms of impurities present in PG. EDS mapping was used to study the distribution of elements, especially the main impurity elements

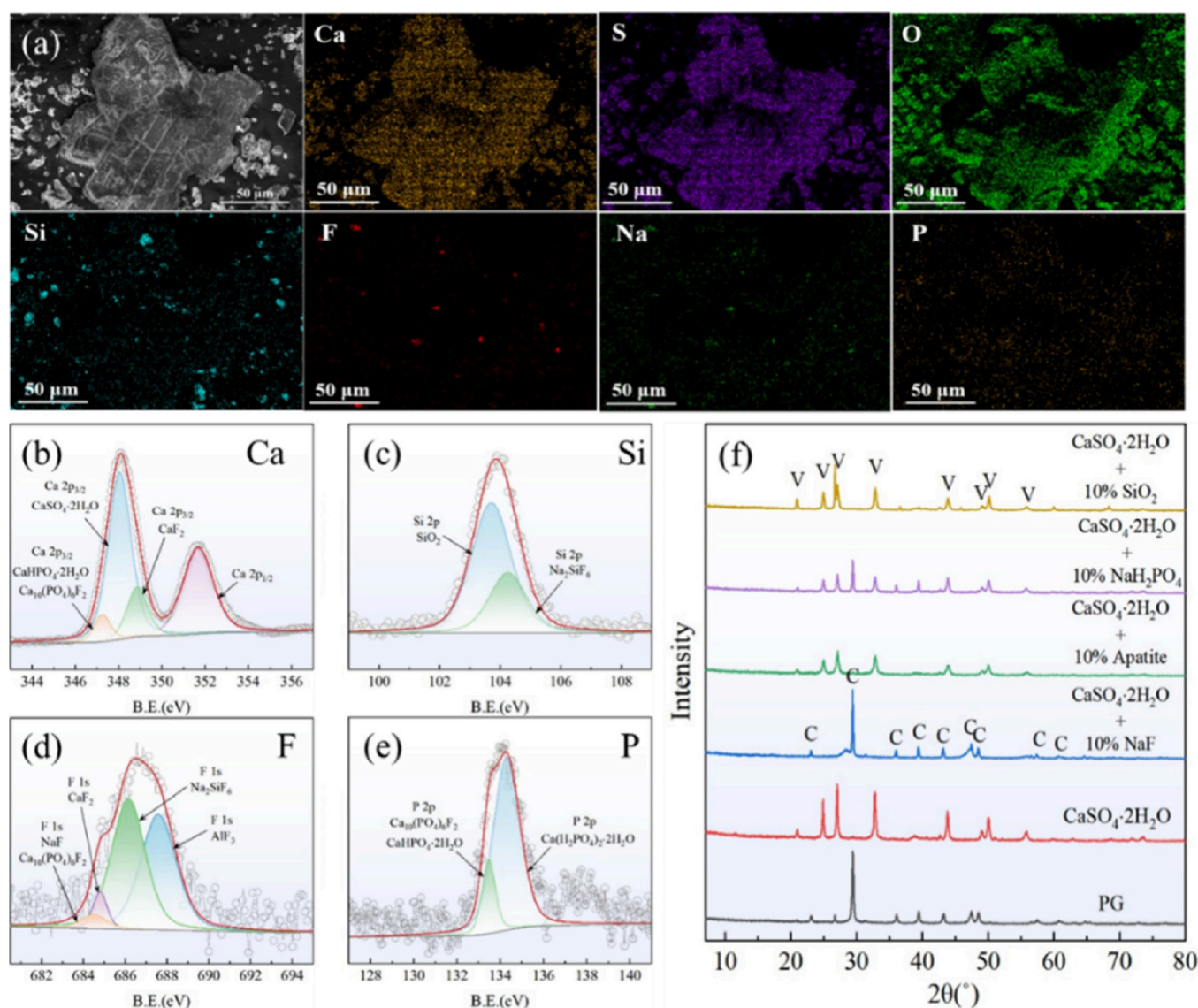


Figure 4. (a) SEM-EDS images of untreated PG; high-resolution XPS spectra of untreated PG: (b) Ca, (c) Si, (d) F, (e) P; (f) XRD patterns of the carbonated products under the addition of different types of impurities.

Si, P, and F, and the results are shown in Figure 4a. The Ca, S, and O elements in $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, which is the main component of PG, are evenly distributed on the surface of PG particles with overlapping areas and clear interfaces. Conversely, Si, F, and Na elements are more sporadically dispersed both on and around the particle surfaces, showing partial overlap in their distribution areas. It is worth noting that Si element displays significant enrichment on the irregular small particles near the large PG particles. The P element, present in relatively low concentrations, exhibits a distribution pattern that is largely consistent with that of Ca.

Moreover, the present forms of major impurity elements in PG were further studied by high-resolution XPS to distinguish these elemental binding states of impurities, and the results are shown in Figure 4b–e. The Ca spectrum profile in Figure 4b exhibits characteristic 2p orbital splitting patterns.³⁹ A dominant spectral signature at 347.7 eV aligns with the Ca 2p_{3/2} electronic state in $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, accompanied by the peak appearing at 351.2 eV corresponding to Ca 2p_{1/2}. The peak at 347.2 eV indicates the presence of $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$ and $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$, while the peak at 349.2 eV confirms the presence of CaF_2 .

Si chemical binding state analysis presented in Figure 4c distinguishes two principal components.⁴⁰ The predominant peak at 103.1 eV corresponds to SiO_2 , confirming quartz as the major silicone impurity present in PG. A secondary peak at 104.3 eV suggests the presence of Na_2SiF_6 , potentially arising from acid treatment processes of phosphate rock. F spectrum mapping in Figure 4d reveals multiple binding forms.⁴¹ The peak at 684.6 eV originates from F 1s in $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$, indicating that there are residual mineral phases that come from wet-process phosphoric acid production. Spectral decomposition also identifies four other distinct fluorine impurities: NaF peak at 684.5 eV, CaF_2 peak at 684.8 eV, Na_2SiF_6 peak at 686.0 eV, and AlF_3 peak at 687.8 eV. The high-resolution spectrum of P 2p, as shown in Figure 4e, demonstrates a bimodal distribution.⁴² The dominant signature centered at 134.1 eV is characteristic of $\text{Ca}(\text{H}_2\text{PO}_4)_2 \cdot \text{H}_2\text{O}$. Concurrently, the adjacent peak at 133.4 eV confirms the simultaneous presence of $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$ and $\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$. These spectral characteristics collectively confirm incomplete apatite decomposition during industrial processing.

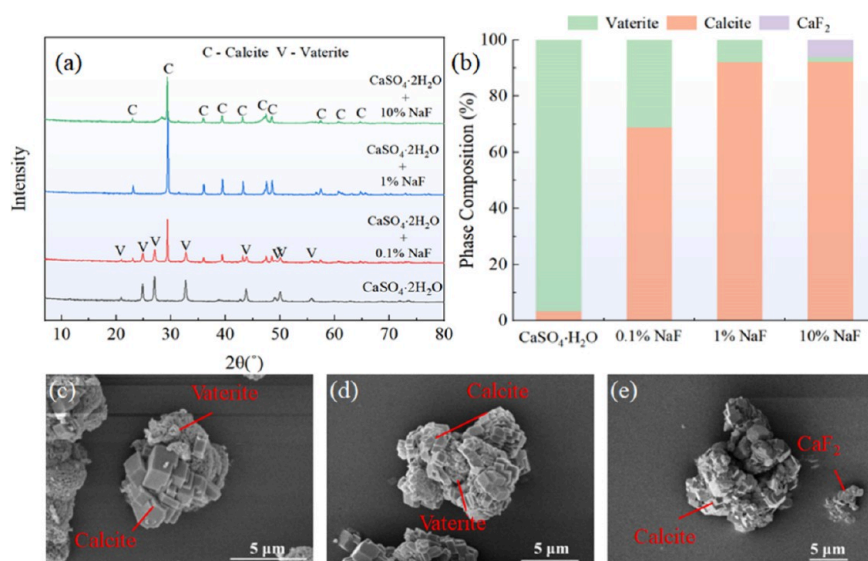


Figure 5. (a) XRD patterns of the carbonated products under different NaF additions and (b) distribution of individual substances within the carbonated products. SEM images of the carbonated products with an addition of (c) 0.1%, (d) 1%, and (e) 10% NaF.

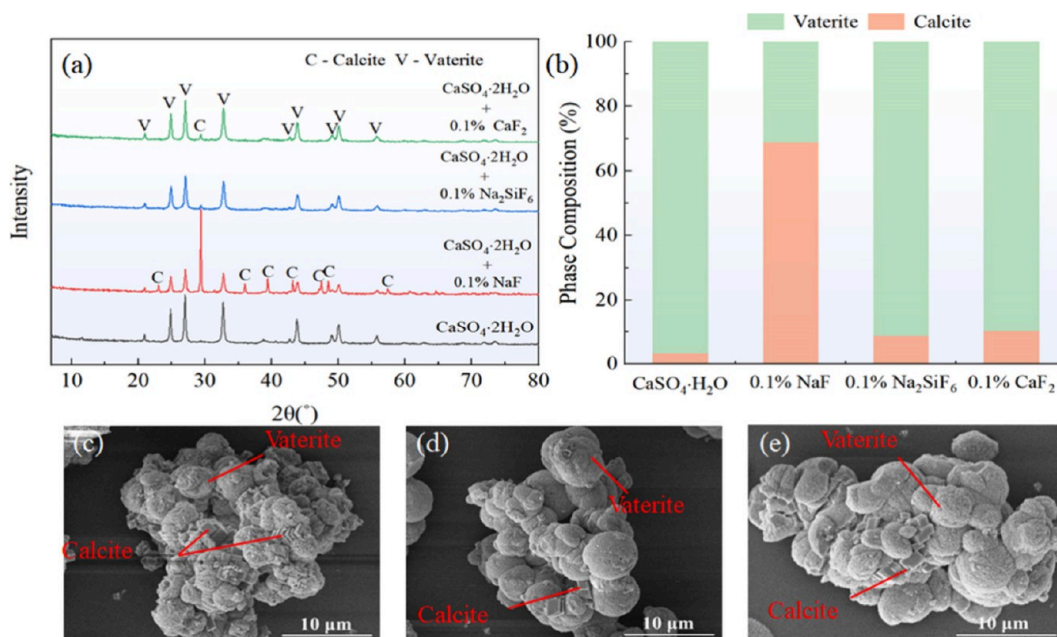


Figure 6. (a) XRD patterns of the carbonated products under the different types of fluorine impurities additions; (b) distribution of individual substances within the carbonated products; SEM images of the carbonated products with an addition of 0.1% (c) NaF, (d) Na_2SiF_6 , and (e) CaF_2 .

In summary, the main impurity components in PG can be roughly divided into three categories: quartz, apatite, and fluorine. In order to investigate the influence of these three types of impurities on the preparation of vaterite, 10 wt % SiO_2 , NaH_2PO_4 , apatite, and NaF were, respectively, added to pure $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ reagent. With an increase in the concentration of individual impurity categories, their influence on calcium carbonate crystal phases becomes more pronounced, thus facilitating the elucidation of underlying mechanisms. From the experimental results shown in Figure 4f, it can be seen that the addition of SiO_2 or apatite did not alter the crystal type of the CaCO_3 product, which remained vaterite. In contrast, the use of NaF as an additive resulted in the complete conversion of vaterite CaCO_3 to calcite. When adding 10% NaH_2PO_4 , the primary product of gypsum mineralization for CO_2 fixation

remains vaterite. Although minor calcite formation occurs, the influence of soluble phosphorus is significantly less pronounced than that of soluble fluoride on vaterite crystallization. These findings suggest that the fluorine impurities present in untreated PG are the primary reason for the formation of calcite exclusively during the carbonation process.

3.2. Impact of Fluorine Content on the Preparation of Vaterite

Based on the aforementioned analysis, it is found that fluoride impurities in PG are the main factor affecting the vaterite generation during the mineral carbonation process. To investigate the impact of varying concentrations of fluorine on vaterite-type CaCO_3 , a series of mineral carbonation experiments were designed using $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, with the addition of NaF at concentrations of 0.1, 1, and 10%, respectively. The

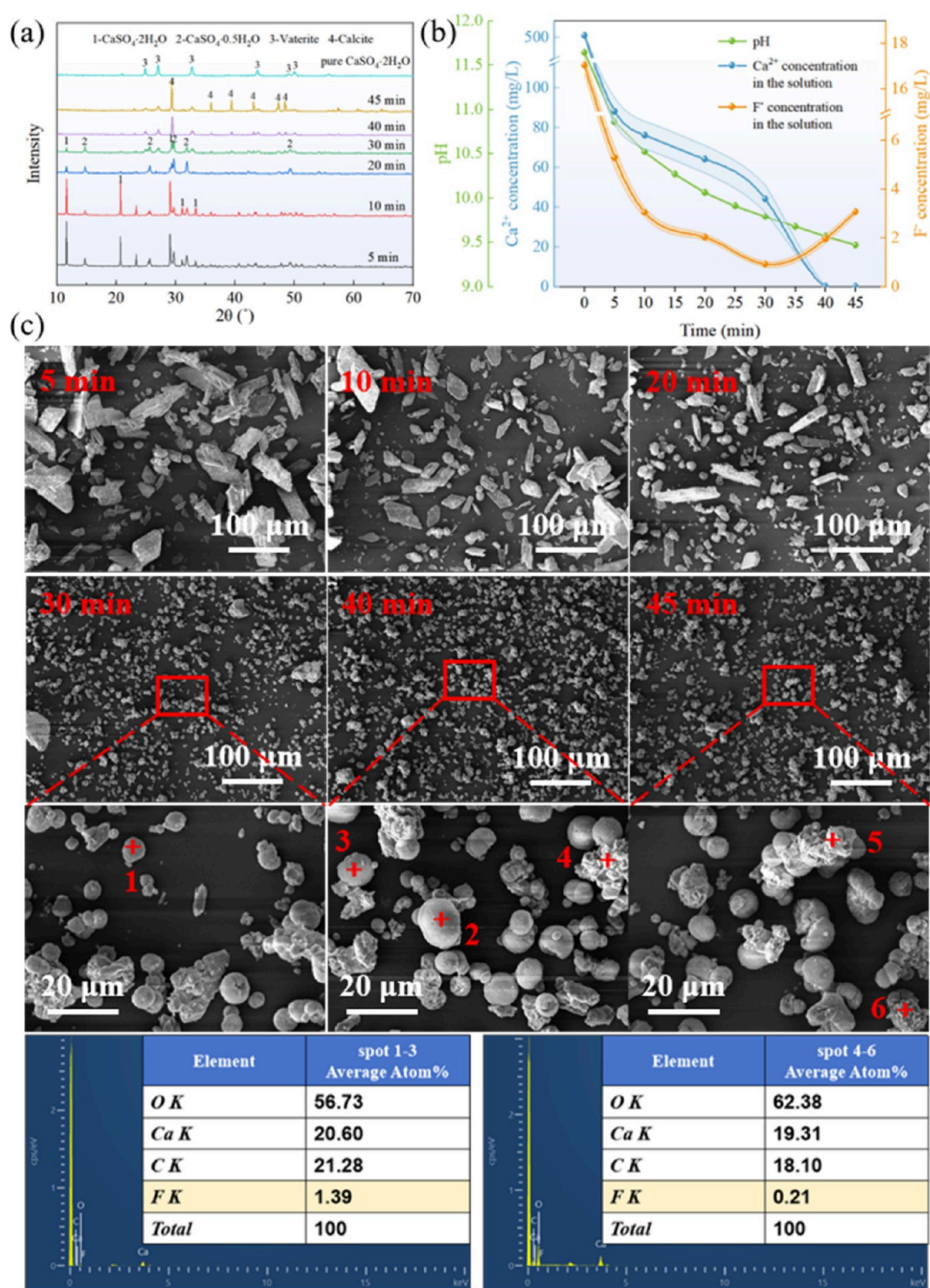


Figure 7. Changes in the (a) phase structure of solids and (b) ion concentrations in the system after adding 0.1% NaF at different time intervals. (c) Transformation process of CaCO_3 morphology.

experiments with impurity additions increased by orders of magnitude were designed to amplify their effects. A comparative analysis was conducted on its carbonated products, and the results are shown in Figure 5. Figure 5a displays the results of XRD patterns for the carbonated products obtained with different NaF additions. It is revealed that a 0.1% addition of NaF results in the appearance of calcite. When the addition of NaF exceeds 1%, the carbonated products are almost entirely converted into calcite-type CaCO_3 . Additionally, a minor

quantity of CaF_2 was identified in the carbonated product when the amount of NaF reached 10%.

Figure 5b exhibits the distribution of individual substances within the carbonated products. It is evident from the graph that at a 0.1% addition of NaF, the carbonated products comprise 68.8% calcite and 31.2% vaterite. Upon increasing the addition of NaF to 1%, the calcite content within the carbonated products increases to 92.1%, while the vaterite content decreases to 7.9%. As the addition of NaF escalates to 10%, the content of calcite

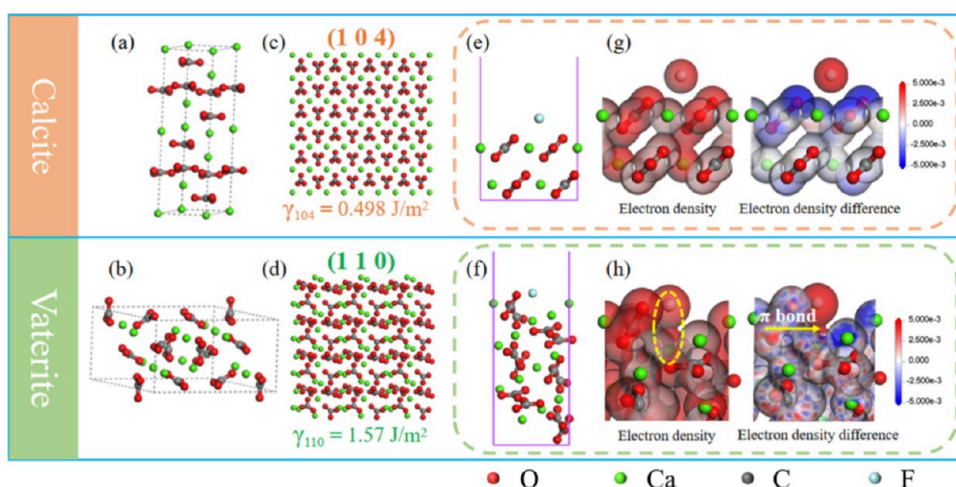


Figure 8. (a, b) Crystal structure of calcite and vaterite; (c, d) surface energy for different facets; (e, f) adsorption configurations for different facets; and (g, h) electron density and electron density difference for different adsorption configurations.

remains at 92.2%; however, the content of vaterite further decreases to 1.6%. At the same time, the carbonated products also contain 6.2% of CaF_2 . Figure 5c depicts the morphological changes of the carbonated products with the increase in NaF addition. The porosity of the vaterite increases as the addition of NaF is 0.1%, and a large number of blocky calcite particles begin to appear on its surface. When the amount of NaF is increased to 1%, calcite further grows and aggregates, surrounding the untransformed vaterite. At this point, the aggregation pattern of calcite still tends to be spherical. When the addition of NaF reaches 10%, calcite growth spreads outward, with vaterite no longer being observed. It can be seen that a very small amount of F content would lead to a large number of vaterite-type CaCO_3 to convert into calcite-type CaCO_3 .

3.3. Impact of Fluorine Types on the Preparation of Vaterite

Fluorine impurities in PG can be divided into three categories according to their different solubility: easy to dissolve, slightly soluble, and difficult to dissolve. The representative components are NaF, Na_2SiF_6 and CaF_2 , respectively. In the experiment of pure $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ mineral carbonation, 0.1% of these three fluorine impurities is individually added to compare their influence on the products of vaterite. In Figure 6a,b, the distribution of each substance in the carbonated product is depicted. The slightly soluble Na_2SiF_6 and insoluble CaF_2 have little effect on the formation of vaterite. In their respective carbonated products, the calcite content is approximately 10%, which is significantly lower than the 68.8% calcite content observed in the carbonated products when 0.1% soluble NaF is added. Compared to the 97% vaterite content of the carbonated products without any impurities, the conversion of vaterite in these cases is relatively minor.

Figure 6c displays the morphological changes of the carbonated products under different types of fluorine impurity additions. The carbonation products obtained by adding slightly soluble Na_2SiF_6 and insoluble CaF_2 exhibit similar characteristics. The vaterite particles retain a relatively intact structure with minimal calcite precipitation observed on their surfaces. Conversely, the addition of easily soluble NaF severely disrupts the vaterite crystal structure, causing its surface to become less dense. Simultaneously, a more substantial presence of calcite structures is noted. Therefore, among the various fluorine-containing impurities contained in PG, the easily soluble

fluorine-containing impurities exert the most serious damage to the structure of vaterite.

3.4. Material Transformation Process in the System

To further investigate the influence of soluble fluoride impurities on the preparation of vaterite by PG direct mineralization, we monitored the transformation of solid products and the variation of ion concentrations in the solution during the reaction with the addition of 0.1% NaF, with the results shown in Figure 7. As can be seen from Figure 7a, a significant amount of vaterite is generated from the conversion of CaSO_4 at 30 min. By 40 min, CaSO_4 is almost completely converted into CaCO_3 , and a partial transformation of vaterite into calcite occurs. When the reaction proceeded to 45 min, the content of calcite further increased. This indicates that calcite formation occurs later than vaterite formation. In contrast, under identical conditions without the addition of NaF, the product at 45 min does not contain calcite. The variation of ion concentrations in the solution during the reaction is shown in Figure 7b. With the continuous introduction of CO_2 , the pH of the solution, as well as the concentrations of Ca^{2+} and F^- , continued to decrease. At 40 min of reaction, due to the complete conversion of CaSO_4 , the Ca^{2+} concentration in the solution approached zero. Meanwhile, accompanying the transformation of vaterite into calcite, the F^- concentration in the solution showed a certain degree of rebound. The SEM-EDS images of the solid products during the reaction are shown in Figure 7c, and vaterite forms even in the early stages of the reaction. Consistent with the XRD results, vaterite began to appear extensively when the reaction reached 30 min. Between 30 and 45 min, vaterite was converted into calcite. Meanwhile, the fluorine content on the surface of vaterite was higher than that of calcite, which also explains the rebound in the F^- concentration observed in the solution during this stage. In summary, vaterite exhibits a certain adsorption capacity for free F^- in the solution, which is also the main reason for the transformation from vaterite to calcite during the middle and late stages of the reaction.

3.5. DFT Calculations

As shown in Figure 8, calcite belongs to the cubic crystal system, and the coordination number of Ca^{2+} is 8.⁴³ The basic phase of vaterite has a monoclinic lattice possessing pseudohexagonal symmetry, with each Ca^{2+} ion being surrounded by at least 12 CO_3^{2-} ions.⁴⁴ The different coordination numbers indicate that a

lower $\text{Ca}^{2+}/\text{CO}_3^{2-}$ ratio favors the formation of vaterite. We calculated the surface energy of the (104) facet of calcite and the (110) facet of vaterite using the CASTEP module of the MS software. As shown in Figure 8a, the (104) facet of calcite, with a low surface energy of 0.498 J/m^2 , is a thermodynamically stable facet. While the (110) facet exhibits the highest surface energy of 1.57 J/m^2 , it is thermodynamically unstable. The adsorption configurations of F^- on the surface of calcite and vaterite were constructed and are shown in Figure 8. The first atomic layer of the (104) facet of calcite and (110) facet of vaterite is mainly composed of exposed Ca^{2+} , which provides the adsorption site for F^- . Based on these adsorption configurations, the adsorption energy and electron density were calculated.

Interfacial adsorption mechanisms can be categorized based on interaction strength and bonding characteristics. Physisorption, governed predominantly by van der Waals interactions between the adsorbent and mineral surfaces, typically exhibits binding energies below 42 kJ/mol . In contrast, chemisorption involves covalent/ionic bond formation through electron transfer processes or surface complexation reactions, with energy magnitudes spanning $40\text{--}400 \text{ kJ/mol}$.⁴⁵ This energy threshold differentiation provides critical insights into the adsorption mechanism identification and interfacial reactivity assessments. As can be seen from Table 2, the adsorption

Table 2. Adsorption Energy of F^- on Different Surfaces of Calcite and Vaterite

surface	adsorbate	E_{ads} (kJ/mol)
(104) facet of calcite	F^-	24.16
(110) facet of vaterite		141.64

energies of F^- on the (104) facet of calcite and (110) facet of vaterite are 24.16 and 141.64 kJ/mol , respectively. It shows that the adsorption of F^- on the (104) facet of calcite and (110) facet of vaterite is mainly governed by the van der Waals force and chemical adsorption, respectively. Figure 8g,h illustrates the electron density and electron density difference in configurations after F^- adsorption. On the (104) facet of calcite, the electron of F^- does not overlap with other atomic orbitals, and there are no gain and loss of electrons around it. Contrarily, as shown in Figure 8h, the electrons of F^- and $\text{Ca}^{2+}(\text{surf})$ overlap in the dotted area and form a π bond. Therefore, F^- will be adsorbed onto the (110) facet of vaterite but has no significant effect on the (104) facet of calcite, suggesting that F^- adsorption on vaterite (110) surfaces involves strong chemical interactions, destabilizing the structure and facilitating its transformation to calcite.

3.6. Preparation of Vaterite from Flotation PG by Direct Mineralization

The flotation process of PG can remove most of the soluble fluorine in PG. Hence, flotation PG was used instead of pure reagent to verify the influence of fluorine on the content of vaterite in PG carbonated products. The chemical composition of PG after flotation is shown in Table 3. Compared with untreated PG, the contents of SiO_2 and F are effectively reduced,

while the P content has little change. Among them, the F content decreased from 0.62 to 0.13% and the SiO_2 content decreased from 5.28% to less than 1%. The component of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ in flotation PG is increased to 97.57% due to the decrease in SiO_2 content. The distribution of elements on the surface of PG after flotation is shown in Figure 9. It indicates a decrease in the presence of irregular fine particles around PG particles. The distribution area of Si and F is significantly reduced relative to untreated PG. In addition, the flotation process results in a smoother surface on the PG particles. The results indicate that flotation can efficiently remove most of the SiO_2 and soluble fluorine impurities, thereby promoting the formation of a greater amount of vaterite.

As depicted in Figure 10, the distribution of various substances in the carbonated products derived from untreated PG and flotation-treated PG is compared. This is consistent with the findings from the experiments using pure reagents, where the vaterite content in these carbonated products rises to over 70% when the fluorine content in PG is reduced to about 0.1%. It is worth noting that the removal of colorants, such as SiO_2 and Fe_2O_3 , has significantly enhanced the whiteness of the carbonated products obtained from flotation PG. Moreover, the resulting vaterite crystals display a well-defined structure and a smooth surface, suggesting that vaterite with an intact structure can be synthesized from PG directly after the flotation treatment.

3.7. Influence Mechanism

Based on the above analysis, the influence mechanism of fluorine on the preparation of vaterite by PG direct mineralization can be depicted in Figure 11. Due to the low solubility of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ at room temperature, the initial concentration of Ca^{2+} in the system is low. When CO_2 is introduced into the ammonia medium, CO_2 reacts rapidly with ammonia to produce a large amount of CO_3^{2-} . At this point, the lower $\text{Ca}^{2+}/\text{CO}_3^{2-}$ ratio facilitates the formation of vaterite. However, the system also contains a certain amount of F^- , which is adsorbed onto the surface of vaterite crystals, disrupting their structural stability. Ultimately, vaterite will redissolve and form calcite crystals. The surface of the calcite crystals exhibits a lower adsorption effect for F^- and has a more stable structure, allowing it to exist stably in a fluorine-containing system. The midproduct obtained in the PG direct mineralization process is shown in Figure 11c. The morphological evolution involving the disintegration of spherical aggregates and concomitant formation of cubic particulates provides direct visual evidence, confirming the phase transformation from vaterite to calcite within this system.

4. CONCLUSIONS

This study systematically elucidated the mechanism by which fluorine impurities impede vaterite formation during PG direct mineralization. The main conclusions are as follows:

- (1) Soluble fluorine species, particularly NaF, act as a dominant inhibitor of vaterite, inducing rapid phase transformation to calcite even at a trace concentration of

Table 3. Chemical Composition of Flotation PG (wt %)

CaO	SO_3	Al_2O_3	SiO_2	P_2O_5	Fe_2O_3	F	H_2O^a	others
32.84	44.31	0.339	0.662	0.769	0.093	0.13	20.42	0.437

^aThe H_2O content specified here denotes the water of crystallization.

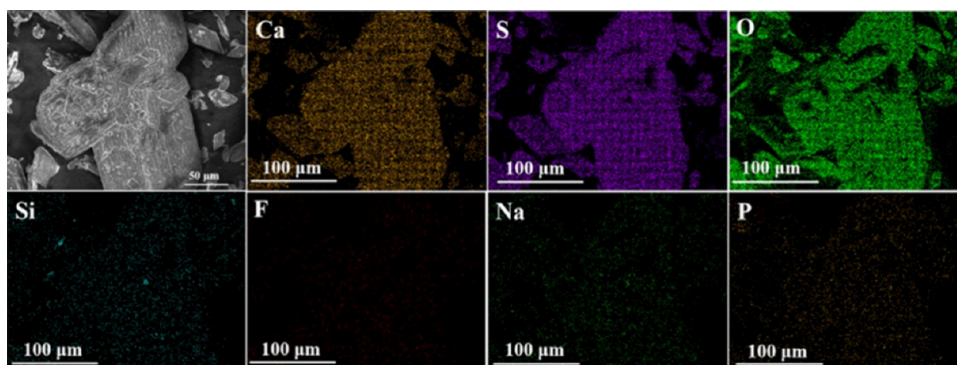


Figure 9. SEM-EDS images of flotation PG.

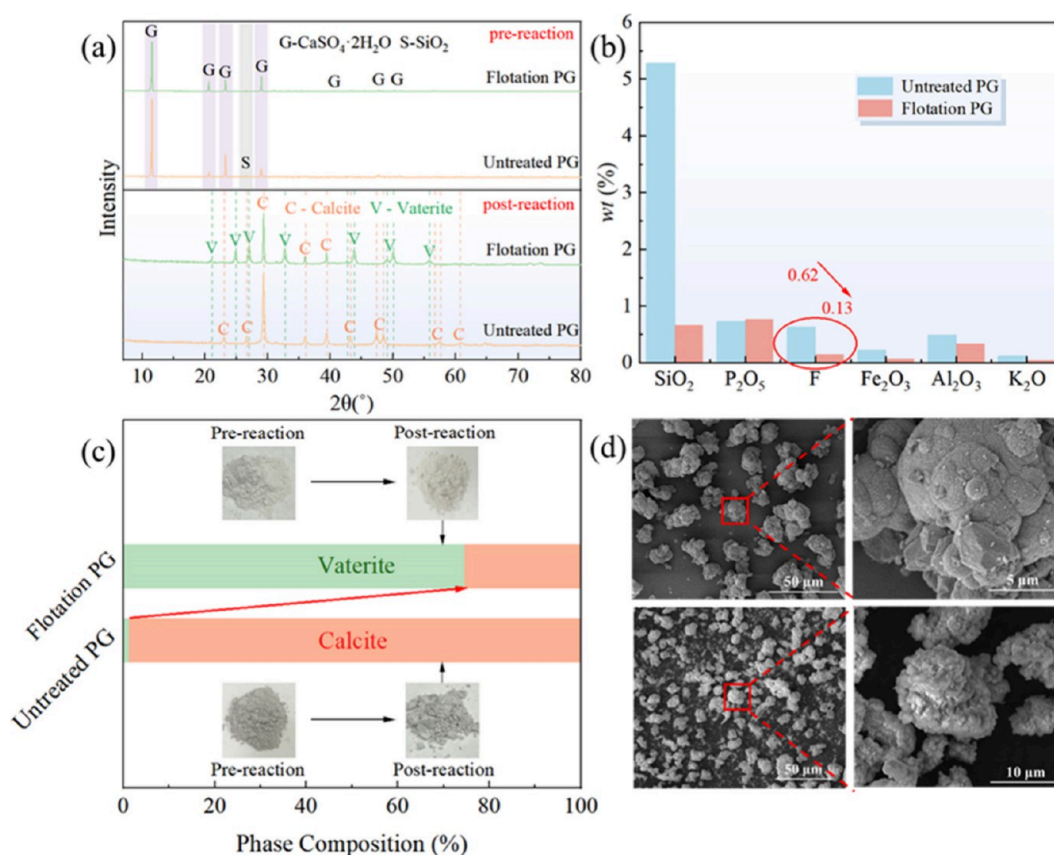


Figure 10. (a) XRD pattern picture; (b) main chemical composition of flotation PG and untreated PG; (c) distribution of individual substances within the carbonated products; and (d) SEM images of the carbonated products from flotation PG and untreated PG.

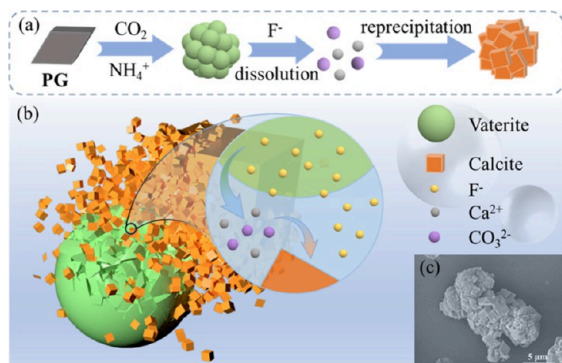


Figure 11. (a–c) Schematic of the influence mechanism of fluorine on the preparation of vaterite by phosphogypsum direct mineralization.

0.1 wt %. In contrast, slightly soluble and insoluble fluorides (e.g., CaF₂, Na₂SiF₆) exhibited negligible effects.

- (2) DFT calculations revealed that F⁻ preferentially adsorbs onto vaterite (110) surfaces via chemical interactions. Conversely, F⁻ adsorption on calcite (104) surfaces was weak, exhibiting negligible influence on the calcite stability.
- (3) Flotation pretreatment was used to remove soluble fluorine from PG, reducing its content from 0.62% to 0.13%, which enabled the direct synthesis of vaterite-rich CaCO₃ (70% vaterite) under identical conditions. Besides, the removal of colorants like SiO₂ further improved product whiteness and crystallinity, enhancing the value of the vaterite product.

- (4) During the mineralization of PG for CO₂ sequestration, a low Ca²⁺/CO₃²⁻ ratio in the system facilitates vaterite formation. However, the system also contains a certain amount of F⁻, which is adsorbed onto the surface of the vaterite crystals, destabilizing its lattice and accelerating recrystallization into calcite.

These insights highlight the importance of fluorine speciation and concentration control in the synthesis of vaterite. The findings not only advance the understanding of impurity-mediated CaCO₃ polymorphism crystallization mechanisms but also offer a scalable strategy for converting PG waste to high-value vaterite.

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Notes

The authors declare no competing financial interest.

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